

☒ Introduction to IDS and IPS – In-Depth Theory

☒ 1. IDS / IPS: Overview

☒ IDS (Intrusion Detection System)

An **IDS** is a monitoring system that **detects suspicious activity or policy violations** and logs or alerts administrators. It is a **passive system** — it does not block traffic but observes and notifies.

☒ IPS (Intrusion Prevention System)

An **IPS** builds on IDS by not just detecting, but also **taking actions** like:

- Blocking malicious traffic
- Resetting connections
- Reconfiguring firewalls

⌘ Key Difference:

Feature	IDS	IPS
Nature	Passive	Active
Action	Alerts only	Alerts + Blocks
Placement	Often outside inline path	Inline (directly in traffic path)

⌘ 2. Types of Attacks Detected by IDS/IPS

IDS/IPS can detect a wide variety of attacks. They fall into the following categories:

⌘ 1. Network Attacks

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DoS / DDoS – Overwhelm target with traffic

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- Port Scans** – Reconnaissance to find open ports
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- Spoofing** – Fake source IP addresses

☒ 2. Exploits

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- Buffer Overflow** – Overwriting memory
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- SQL Injection** – Injecting malicious SQL commands
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- Cross-Site Scripting (XSS)** – Injecting malicious scripts

☒ 3. Malware

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Trojans, Worms, Ransomware, etc.

☒ 4. Insider Threats

- Misuse of privileges by legitimate users

☒ 3. IDS Detection Techniques

IDS systems detect attacks using one or more of the following techniques:

☒ Signature-Based Detection

- Matches traffic against known attack patterns
- Low false positives
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Cannot detect unknown threats (zero-day attacks)

☒ **Anomaly-Based Detection**

- Creates a baseline of "normal" activity
- Flags deviations
- Can detect unknown threats
- Higher false positives

☒ **Heuristic-Based Detection**

- Uses behavioral analysis and rule-based logic
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Advanced, less common

❏ 4. Types of IDS

Type	Description	Placement
NIDS – Network-based	Monitors network traffic (e.g., Snort)	Deployed at network boundaries
HIDS – Host-based	Monitors host system logs and behavior	Installed on individual systems
Hybrid IDS	Combines NIDS + HIDS	For layered visibility

❏ 5. Security Events

A **security event** is any observable occurrence in a system or network. Not all security events are threats, but:

- A **security incident** is a series of related events representing a threat
- IDS/IPS help **distinguish benign events from malicious ones**

Examples of security events:

- Multiple failed login attempts
 - Unexpected system reboots
 - High CPU usage on idle servers
 - Connection to known malicious IPs
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⌘ 6. Vulnerability, Design Flaws & Implementation Weaknesses

⌘ Vulnerability

A weakness in software or hardware that can be exploited to compromise a system.

⌘ Design Flaw

- Architectural mistake (e.g., trusting unauthenticated input)
- Harder to detect, often systemic

⌘ Implementation Bug

- Code-level flaw (e.g., improper input validation)
- More common, patchable

⊠ IDS/IPS may not detect:

- Encrypted payloads (if not configured properly)
- New attack types without updated signatures
- Malware using obfuscation techniques

⊠ Proper design, frequent signature updates, and contextual threat intelligence improve detection accuracy.